AI for Redistricting — Spring 2026

Gerrymandering Analysis of Massachusetts Congressional Districts

2024 Presidential & Senate Elections | ReCom Ensemble Analysis

University of San Francisco • CS 464

Massachusetts Context

9
Congressional Districts

7.03M
Total Population (2020)

100%
Dem-held seats (current)

2023
Current map enacted

Map Creation

The map was enacted in 2023 by the Democratic-controlled state legislature with no independent redistricting commission.

Partisan Landscape

Democrats hold all 9 congressional seats, with Harris winning ~64% of the statewide presidential vote in 2024.

Legal Landscape

The map has not faced legal challenges, though its entirely Democratic delegation raises questions about partisan gerrymandering.

Racial Landscape

Massachusetts is majority white (~71%), with minority populations concentrated in Boston, anchoring District 7 as a majority-minority district.

Data & Methods

Data Sources

1 **2020 Census Blocks**

107,278 census blocks with total population (P0010001). Used to aggregate population to precinct level.

2 **MA County Shapefile**

14 counties. Used to nest precincts within counties during MAUP repair.

3 **2024 General Election Precincts**

2,383 precincts with vote totals for Presidential (Harris/Trump) and Senate races.

MAUP Data Cleaning Pipeline

1 **Reproject**

Aligned all three shapefiles for accurate spatial operations.

2 **smart_repair**

Fixed 519 overlaps, 77 holes, and invalid geometries in election precincts, nesting within county boundaries.

3 **Rook → Queen**

Converted adjacencies < 30m to point adjacencies so GerryChain treats them as proper neighbors.

4 **Assign Population**

Used maup.assign() to map 107k census blocks → 2,383 precincts and sum population.

5 **Validation**

maup.doctor() returns True. Block total = Precinct total = 7,029,917.

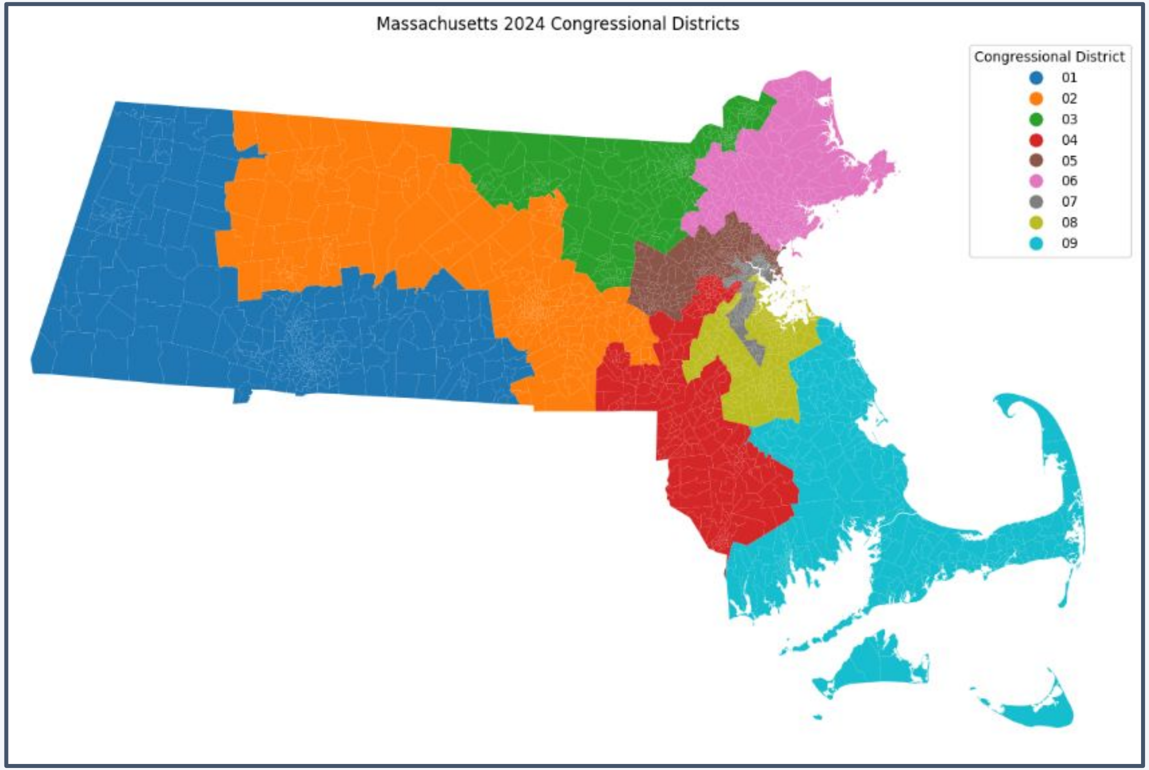
Ensemble Method: ReCom

How ReCom Works

1. Pick two adjacent districts at random.
2. Merge them into one region.
3. Build a random spanning tree of the merged region.
4. Cut one edge of the spanning tree to create two new valid districts satisfying population constraints.
5. Repeat 20000 times to generate the ensemble.

Parameter Choices	
Steps per chain	**20000**
Number of chains	**2 (Pres + Senate)**
Population tolerance	**±2%**
Node repeats	**2**
Compactness bound	**≤ 2× initial cut edges**

The Actual Massachusetts Map



9 districts
All Democrat-held

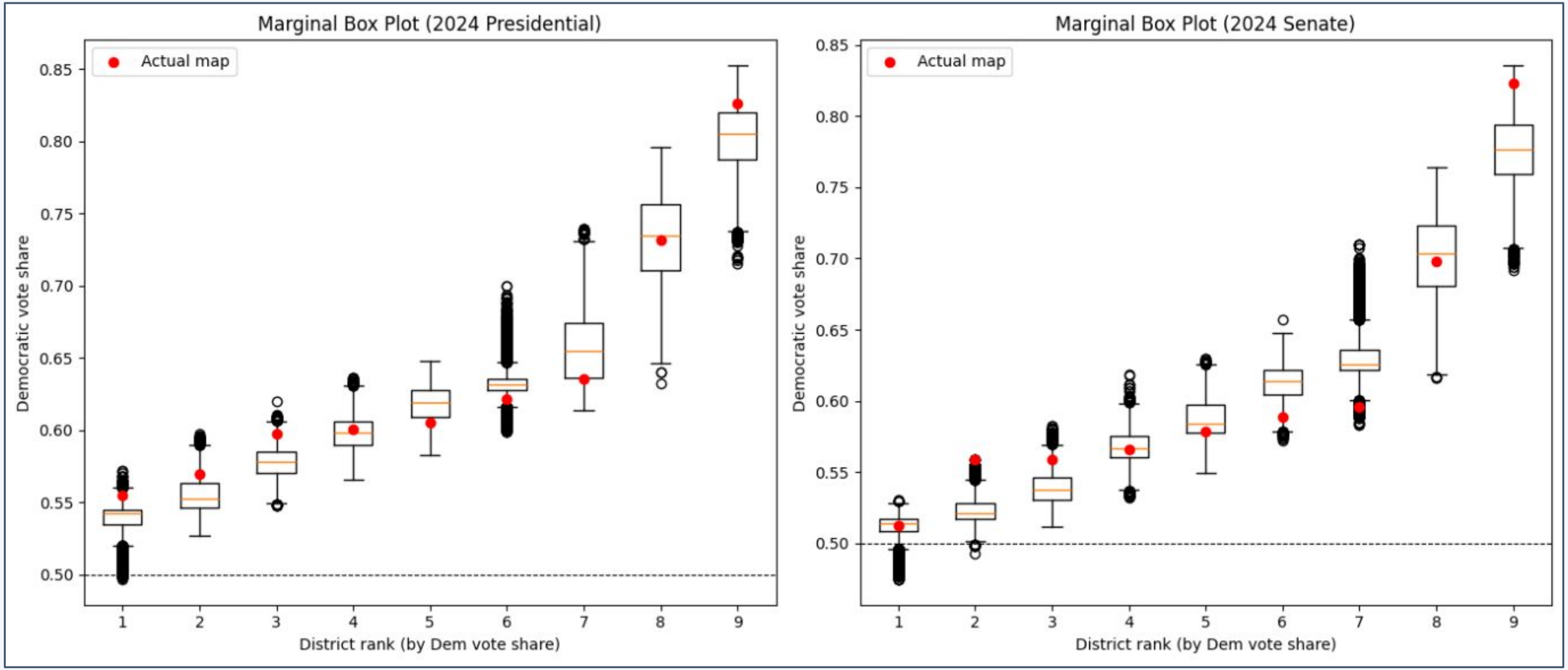
659
Cut edges in actual map

-0.24
Efficiency Gap (Pres)

-0.30
Efficiency Gap (Senate)

Outlier Analysis: Histograms

Red line = actual MA map value. Each histogram shows where the MA map falls relative to 20000 randomly generated alternative maps.



Key finding: The actual map's Efficiency Gap sits in the extreme left tail of both distributions, and all 9 Democrat wins is a rare outcome in the ensemble.

Understanding the Efficiency Gap

What is the Efficiency Gap?

Measures wasted votes — all losing-party votes are wasted, plus any winning-party votes beyond the 50%+1 threshold needed to win.

EG = (Wasted Dem − Wasted Rep) / Total Votes

Negative EG → map favors Democrats

Positive EG → map favors Republicans

EG Results

Election	Ensemble Mean	Actual Map
Pres 2024	-0.12	-0.24
Senate 2024	-0.14	-0.30

Compactness: Polsby-Popper Score

Polsby-Popper

$$PP = 4\pi \times Area / Perimeter^2$$

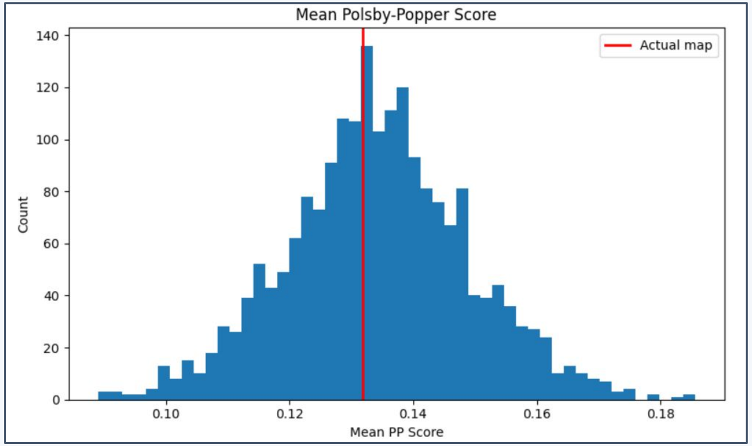
Score of 1.0 = perfect circle (most compact)

Score near 0 = very irregular or fragmented shape

The actual MA map has more cut edges (659) vs ensemble mean (~480), suggesting deliberately fragmented boundaries.

Actual Map PP Scores

D01	0.2797
D02	0.2119
D03	0.3264
D04	0.2341
D05	0.3876
D06	0.2654
D07	0.1823
D08	0.1934
D09	0.2512



Conclusions

Extreme Efficiency Gap

The actual MA map shows an EG of −24% (Pres) and −30% (Senate), both in the extreme left tail of the ensemble — far more Democratic-favoring than nearly all 2,000 randomly generated maps.

All 9 Districts Democrat

Winning all 9 districts is a rare outcome in the ensemble, suggesting the map is drawn in a way that maximizes Democratic seats.

More Fragmented Borders

659 cut edges vs ensemble mean of ~480 indicates the map has more fragmented district boundaries than typical — consistent with deliberate line-drawing.

District 9 Packing

Marginal box plots show District 9 is packed with Democratic votes beyond what random maps typically produce.